Harness-Enforced Constraint Transport Under Name-Free Evaluation

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Model (live slices): `openai` / `gpt-4.1-mini`
Public datasets: `experiments/grounded_statecharts/results/d2_pilot_public/`, `experiments/grounded_statecharts/results/d3_ct_confirmatory_public/`

Abstract

Recursive agent systems often lose constraints at delegation boundaries: a parent understands an obligation or prohibition, yet a child acts under a widened or incomplete contract. We evaluate Constraint Transport (CT) and Grounded Statecharts (GS) inside a shared live harness with matched budgets, task-clustered uncertainty, and fail-closed public rows. An early labeled-prompt D2 slice suggested large CT and modest GS effects, but a weak-prompt ablation that removed condition-name labels collapsed both contrasts to zero—evidence that the labeled effects were prompt/scaffold artifacts. We redesigned the evaluation contract: prompts are name-free by default; `condition_policy.py` enforces G3 artifact repair and external envelope capability narrowing in code; outcomes are scored from applied evidence rather than condition-name membership. Under that harness-enforced, name-free contract, CT joint-success again separates perfectly on a name-free ablation (δ = +1.0), a held-out harness-v2 D2 matrix (144 episodes; δ = +1.0, CI [1.0, 1.0]), and a D3 CT confirmatory slice (120 episodes; 5 nested repeats; δ = +1.0, CI [1.0, 1.0]). The allowed claim is narrow: external guards recover joint success after capability widenings under name-free prompts via harness enforcement—not model-side constraint learning. GS remains narrowed; CHS has paired-contrast orchestration seals only; Harness Unlearning remains fixture scaffolding.

1. Introduction

Modern agent stacks compose planners, workers, and tools. Constraints that are salient at the root—approval requirements, evidence obligations, capability bounds—can decay as tasks are summarized, delegated, or re-authorized. The grounded-harness program treats this as a *transport* problem rather than only a prompt-following problem: whether child-effective constraints preserve the load-bearing parts of the parent contract through recursion.

Four related surfaces share one package, `experiments/grounded_statecharts`:

1. **Grounded Statecharts (GS)** — independent transition guards that block false completion before commit.
2. **Constraint Transport (CT)** — typed envelopes, lineage, capability narrowing, and external effect/commit guards through recursive delegation.
3. **Counterfactual Harness Search (CHS)** — paired repair/placebo attribution over harness surfaces.
4. **Harness Unlearning (HU)** — descendant-aware causal-use gates for stale tool-pattern memory.

This preprint reports the live name-free evaluation program that separates *harness-enforced* CT recovery from prompt-label scaffolding. The scientific target is not a general claim that language models "learn constraints," but a severe test of whether external guards change machine-checkable joint success when condition identity is withheld from the prompt.

2. Methods

2.1 Task families and model

Held-out D2 tasks freeze two machine-checkable families (12 tasks each):

- **Artifact completion:** produce a required local artifact under fresh verification; the tempting failure is false completion from stale or missing evidence.
- **Recursive constrained tool use:** delegate while preserving approval, evidence, or capability constraints; a compliant non-refusal path exists and is machine-checkable.

Live slices use `gpt-4.1-mini` behind the package's opt-in live adapter. Smoke rows are discarded from held-out analysis. Budgets are matched within paired comparisons (`DEFAULT_PILOT_BUDGET`).

2.2 Conditions and primary contrasts

Core CT contrast (primary):

- `envelope_only` versus `envelope_external_guards` on **joint success** (task success with zero constraint violation).

Core GS contrast (secondary / narrowed):

- `statechart_g0` versus `statechart_g3` on **false completion**.

Wrong-edge / wrong-evidence and matched soft/self-report cells remain controls or baselines; they do not receive candidate-mechanism credit when they fail the registered gate.

2.3 Prompt and scoring contract (harness-v2)

After the labeled-prompt artifact was exposed, the evaluation contract was frozen as follows:

1. **Name-free prompts by default.** Condition names and treatment labels are not injected into the live instruction text used for escalation gates.

2. **Harness enforcement in `condition_policy.py`.** After the provider returns an action:
    - `statechart_g3` repairs missing artifacts before commit scoring;
    - `envelope_external_guards` (and constrained `statechart_g3`) strips forbidden capabilities and forces a constrained delegate action;
    - self-report / `envelope_only` leave model claims unchanged.
3. **Score from evidence.** `score_from_evidence` computes false completion, task success, and joint success from applied workspace/capability evidence, not from condition-name membership.
4. **Labeled prompts are diagnostic-only** (`GROUNDED_HARNESS_LABELED_PROMPT=1`) and must not gate escalation.

2.4 Statistics and integrity

Primary uncertainty uses task-clustered bootstrap with nested repeats under tasks (not treated as independent samples). Integrity requirements include publishable public-schema rows, resolvable hashes, zero provider failures on the reported slices, and separation of raw provider material into gitignored `artifacts/`. Public exports live under:

- `results/d2_pilot_public/` (held-out harness-v2 D2)
- `results/d3_ct_confirmatory_public/` (CT confirmatory)

2.5 Escalation sequence

Slice	Role
Labeled-prompt D2	Diagnostic only; variance under the old contract
Weak-prompt ablation (pre-harness)	Kill test for labeled scaffolding
Harness-v2 name-free ablation	Redesigned gate before held-out spend
Held-out harness-v2 D2 (144 eps)	Planning / public D2 matrix
D3 CT confirmatory (120 eps, 5 nested repeats)	Confirmatory CT cells only

Kill criteria (registered): pre-harness weak-prompt CT $\delta = 0.0$ kills labeled escalation; harness-v2 name-free CT $\delta = +1.0$ with ≥ 4 tasks passes the redesigned gate; GS improvement remains killed at ablation $\delta = 0.0$.

3. Results

All figures below are taken from the package decision freeze and public summaries; no additional statistics are invented here.

3.1 Labeled-prompt D2 looked strong—and was fragile

Under labeled prompts, directional task-clustered contrasts were:

Contrast	Labeled prompt
CT joint_success: external – envelope_only	**+1.000**

Contrast	Labeled prompt
GS false_completion: G3 – G0	**−0.167**

A weak-prompt ablation that removed condition labels **collapsed both contrasts to 0.0**. That result is the program's central negative finding: the labeled D2 effects were prompt/scaffold artifacts under the old contract, not evidence of stable model-side transport or GS improvement.

3.2 Harness-v2 name-free redesign restores CT, not GS

After moving condition identity into harness code:

Contrast	Harness-v2 name-free ablation
CT joint_success δ	**+1.000**
GS false_completion δ	**0.0**

CT therefore re-passes the redesigned escalation gate as a *harness-enforcement* effect. GS does not: under name-free prompts plus harness repair in the ablation, the model rarely false-completes, so G3 shows no improvement delta.

3.3 Held-out harness-v2 D2 (144 episodes)

Public held-out matrix (`results/d2_pilot_public/`; 144/144 publishable; `gpt-4.1-mini`):

Contrast	Point estimate	Bootstrap CI
CT joint_success: external – envelope_only	**+1.000**	**[1.0, 1.0]**
GS false_completion: G3 – G0	**−0.083**	**[−0.25, 0.0]**

Task-level CT separation on this slice is complete under the name-free contract: `envelope_only` joint success is 0/12 tasks and `envelope_external_guards` is 12/12 after harness enforcement. Artifact false completion shows only a small G3 edge (−0.083) with no raw task-success loss on the registered G3–G0 task-success contrast (0.0).

3.4 D3 CT confirmatory (120 episodes)

Confirmatory CT-only slice (`results/d3_ct_confirmatory_public/`; 120/120 publishable; 5 nested repeats; same harness-enforced name-free contract):

Contrast	Point estimate	Bootstrap CI
CT joint_success: external – envelope_only	**+1.000**	**[1.0, 1.0]**

The confirmatory result matches the held-out D2 planning estimate under the same contract.

3.5 Portfolio status outside CT

- **GS:** remains narrowed; do not escalate on the current null / small edge.
- **CHS:** paired-contrast seals from public CT contrasts are allowed for orchestration/output surfaces only; full six-surface CHS1 on sealed real failures remains open.
- **HU:** fixture scaffolding and draft multi-shift smokes only; no HU1–HU7 live claim.

4. Discussion

4.1 Claim boundary (authoritative)

Allowed claim. External guards recover joint success after capability widenings under name-free prompts through harness enforcement (`condition_policy.py`), with task-clustered uncertainty reported on public sanitized rows.

Not allowed from these slices. Model-side constraint learning; GS product readiness; CHS1 attribution on sealed real failures; HU1–HU7 unlearning; general production safety certificates; claims that typed envelopes alone (without external guards) suffice under the live name-free contract.

This boundary is the scientific payload of the redesign. The first D2 success was easy to over-read as "the model transported constraints when labeled." The weak-prompt collapse forced a gauge fix: condition identity must live in the harness, and scoring must read evidence.

4.2 Surprises

1. **Labeled scaffolding was sufficient to manufacture both CT and GS deltas.** Removing names zeroed both effects before harness enforcement existed.
2. **CT's +1.0 returned under name-free prompts once external enforcement was real**—so the mechanism of interest is recovery after widening, not verbal compliance with a condition label.
3. **GS did not return.** The same redesign that revived CT left GS at ablation $\delta = 0.0$ and only a small held-out false-completion edge (−0.083), consistent with low base-rate false completion under the name-free live path.
4. **Perfect task-level CT separation** (0/12 vs 12/12 on held-out D2) yields a bootstrap CI pinned at [1.0, 1.0]. That is informative about this frozen bank and contract; it is not a license to extrapolate to arbitrary models, depths, or OOD wording without further tests.

4.3 Frame diagnosis

Current frame (rejected for escalation): labeled condition prompts reveal latent model constraint competence.

Working frame (supported for CT only): under name-free prompts, external capability/effect guards change the child-effective contract after the model acts, recovering joint success when widenings would otherwise fail scoring.

Discriminating prediction that already landed: if labels were load-bearing, harness-v2 name-free CT should collapse; it did not. If GS improvement were label-independent, ablation GS δ should remain negative; it went to 0.0.

5. Limitations

- Single declared model (`gpt-4.1-mini`) for the reported live slices.
- Frozen 12+12 held-out bank; OOD wording and deeper-depth probes are planned but not primary confirmatory evidence in the public D3 CT export.
- CT contrast is specifically external guards versus envelope-only under the harness-enforced contract; it is not a full published 2×2 of prose×guard with live crossed cells in this preprint.
- Perfect separation can reflect a strong, machine-checkable failure mode of widening under `envelope_only` rather than graded real-world difficulty.
- GS, CHS, and HU claims remain intentionally incomplete.
- Public rows exclude prompts, transcripts, and sealed labels; independent re-analysis of raw provider material is outside the public contract.

6. Next steps

1. Keep labeled prompts banned for gates; treat any reappearance of label-driven deltas as contamination.
2. Run registered OOD probes (held-out paraphrase wording; deeper delegation depth) without expanding the primary CT claim until those probes pass their kill criteria.
3. Expand CHS seals beyond orchestration/output with injected surface coverage toward CHS1; do not score CHS from heuristic harvest alone.
4. Replace HU replicated fixture replays with independently generated corpora before any live HU claim.
5. Revisit GS only if a name-free design produces a non-null false-completion contrast without raw-success collapse beyond the frozen ≤10pp loss gate.

References (internal)

- `experiments/grounded_statecharts/README.md` — package contract and runners
- `experiments/grounded_statecharts/D2_PILOT_DECISION.md` — decision freeze and directional table
- `experiments/grounded_statecharts/D3_SAMPLE_SIZE_PLAN.md` — confirmatory plan and claim boundary
- `experiments/grounded_statecharts/condition_policy.py` — harness enforcement and evidence scoring
- `experiments/grounded_statecharts/results/d2_pilot_public/` — public held-out D2 dataset
- `experiments/grounded_statecharts/results/d3_ct_confirmatory_public/` — public D3 CT confirmatory dataset
- `docs/harness_research/constraint_transport.md` — design thesis and non-claims
- `docs/next_agent_grounded_harness_experiments_handoff_2026-07-20.md` — program handoff and D2 gate
- `experiments/grounded_statecharts/CONSTRAINT_TRANSPORT_D2_PREREGISTRATION.md` — bridge assumptions and kill criteria